

Problem

A parallel RL circuit has $R = 4\ \Omega$ and $L = 1\ \text{H}$. The input to the circuit is $i_s(t) = 1.4e^{-t}u(t)$ A. Find the inductor current $i_L(t)$ for all $t > 0$ and assume that $i_L(0) = -1.4$ A.

Step-by-step solution

Step 1 of 7

Consider the value of the resistor, $R = 4\ \Omega$

The value of the inductor, $L = 1\ \text{H}$

The input to the circuit is $i_s(t) = 1.4e^{-t}u(t)$ A

The initial inductor current is $i_L(0) = -1.4$ A

Draw the parallel RL circuit in s -domain.

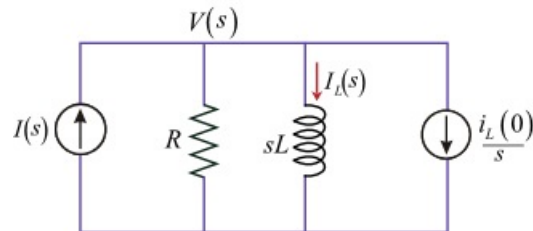


Figure 1

Step 2 of 7

Calculate the Laplace transform of, $i_s(t)$.

$$\begin{aligned}
 \mathcal{L}(i_s(t)) &= I(s) \\
 &= \mathcal{L}(1.4e^{-t}u(t)) \\
 &= 1.4\mathcal{L}(e^{-t}u(t)) \quad \left(\because e^{-t}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+1} \right) \\
 &= \frac{1.4}{s+1}
 \end{aligned}$$

Step 3 of 7

Apply the Kirchhoff's current law to the circuit in Figure 1.

$$I(s) = \frac{V(s)}{R} + I_L(s) + \frac{i_L(0)}{s}$$

Substitute $sLI_L(s)$ for $V(s)$ by virtue of parallel circuit which has the same voltage across the each element.

$$\begin{aligned} I(s) &= \frac{sLI_L(s)}{R} + I_L(s) + \frac{i_L(0)}{s} \\ &= \left(\frac{sL}{R} + 1\right)I_L(s) + \frac{i_L(0)}{s} \end{aligned}$$

$$I_L(s) = \frac{I(s) - \frac{i_L(0)}{s}}{\left(\frac{sL}{R} + 1\right)}$$

Step 4 of 7

Substitute $\frac{1.4}{s+1}$ for $I(s)$, 4Ω for R , 1 H for L , and -1.4 A for $i_L(0)$.

$$\begin{aligned} I_L(s) &= \frac{\frac{1.4}{s+1} - \frac{(-1.4)}{s}}{\left(\frac{s(1)}{4} + 1\right)} \\ &= \frac{\frac{1.4}{s+1} + \frac{1.4}{s}}{\left(\frac{s(1)}{4} + 1\right)} \\ &= \frac{\frac{2.8s + 1.4}{(s+1)s}}{\left(\frac{s+4}{4}\right)} \\ &= \frac{8(1.4s + 0.7)}{(s+4)(s+1)s} \end{aligned}$$

Step 5 of 7

Decompose the $I_L(s)$ into simple terms using the residue method of partial fraction expansion.

$$\begin{aligned} I_L(s) &= \frac{8(1.4s + 0.7)}{(s+4)(s+1)s} \\ &= \frac{A}{(s+4)} + \frac{B}{(s+1)} + \frac{C}{s} \end{aligned}$$

Here A , B , and C are constants.

Calculate the constant value A using the residue method.

$$\begin{aligned} A &= (s+4)I_L(s)\Big|_{s=-4} \\ &= (s+4)\left(\frac{8(1.4s + 0.7)}{(s+4)(s+1)s}\right)\Big|_{s=-4} \\ &= 8\left(\frac{1.4(-4) + 0.7}{(-4+1)(-4)}\right) \\ &= -3.27 \end{aligned}$$

Step 6 of 7

Calculate the constant value B using the residue method.

$$\begin{aligned} B &= (s+1)I_L(s) \Big|_{s=-1} \\ &= (s+1) \left(\frac{8(1.4s+0.7)}{(s+4)(s+1)s} \right) \Big|_{s=-1} \\ &= 8 \left(\frac{1.4(-1)+0.7}{(-1+4)(-1)} \right) \\ &= 1.87 \end{aligned}$$

Calculate the constant value C using the residue method.

$$\begin{aligned} C &= (s)I_L(s) \Big|_{s=0} \\ &= s \left(\frac{8(1.4s+0.7)}{(s+4)(s+1)s} \right) \Big|_{s=0} \\ &= 8 \left(\frac{1.4(0)+0.7}{(0+4)(0+1)} \right) \\ &= 1.4 \end{aligned}$$

Substitute obtained A , B , and C values in equation, $I_L(s)$.

$$I_L(s) = \frac{-3.27}{(s+4)} + \frac{1.87}{(s+1)} + \frac{1.4}{s}$$

Step 7 of 7

Apply the inverse Laplace transform to find $i_L(t)$

$$\begin{aligned} i_L(t) &= \mathcal{L}^{-1}(I(s)) \\ &= \mathcal{L}^{-1} \left[\frac{-3.27}{(s+4)} + \frac{1.87}{(s+1)} + \frac{1.4}{s} \right] \\ &= -3.27 \mathcal{L}^{-1} \left(\frac{1}{s+4} \right) + 1.87 \mathcal{L}^{-1} \left(\frac{1}{s+1} \right) + 1.4 \mathcal{L}^{-1} \left(\frac{1}{s} \right) \quad \left(\because e^{-t}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+1} \right) \\ &= -3.27e^{-4t}u(t) + 1.87e^{-t}u(t) + 1.4u(t) \\ &= (-3.27e^{-4t} + 1.87e^{-t} + 1.4)u(t) \end{aligned}$$

Thus, the inductor current $i_L(t)$ for $t > 0$ is $\boxed{(-3.27e^{-4t} + 1.87e^{-t} + 1.4)u(t)}$.